

Five-Channel Infrared Sensor Array for Autonomous Line Tracking

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1 Introduction

1.1 Aim

The aim of this assignment is to design, program, and interface an autonomous five-sensor line follower mobile robot powered by an Arduino Mega 2560 microcontroller driving two direct current motors through an L298N dual H-bridge motor driver module.

1.2 Overview of Approach

The navigation approach relies on differences in optical surface reflectance between dark and light materials. An array of five infrared emitter and detector pairs is mounted across the front of the mobile robot chassis facing the ground. The infrared diodes illuminate the surface, and the phototransistors detect the intensity of reflected light. White surfaces reflect infrared light effectively while dark surfaces absorb it, generating discrete binary voltage signals across analog pins A0 through A4.

The microcontroller scans these signals, determines the lateral position of the tracked path relative to the robot centerline, and selects corresponding differential drive steering commands. Output signals from the microcontroller govern direct current gear motors via the L298N driver, producing proportional turning, straight progression, or complete stopping when the path is lost.

2 Hardware Used

The experimental robotic platform integrates a high-performance microcontroller board, an optical line tracking module, an integrated dual H-bridge motor driver, two geared direct current motors, a mobile chassis, and electrical interconnects.

2.1 Components

Table 1 lists all physical hardware units used in the mobile robot configuration along with their specific technical roles.

Component	Model / Part	Qty	Purpose
Microcontroller Board	Arduino Mega 2560	1	Main processor and pulse width modulation controller
Line Sensor Array	5-Channel IR Tracker	1	Detects ground reflectance patterns across analog pins A0 through A4
Motor Driver	L298N Dual H-Bridge	1	Powers and controls two direct current gear motors
Drive Motors	5V to 12V DC Gear Motors	2	Differential drive locomotion for left and right wheels
Robot Chassis	Two-wheel drive acrylic chassis	1	Physical body structure supporting components
Jumper Wires	Male-to-male and male-to-female	16	Inter-module electrical wiring

Table 1. Hardware components used in the line follower robot system.

2.2 Wiring Diagram

Figure 1 presents the complete circuit schematic linking the Arduino Mega 2560 microcontroller, the five-channel infrared sensor board, the L298N motor driver, and the direct current gear motors.

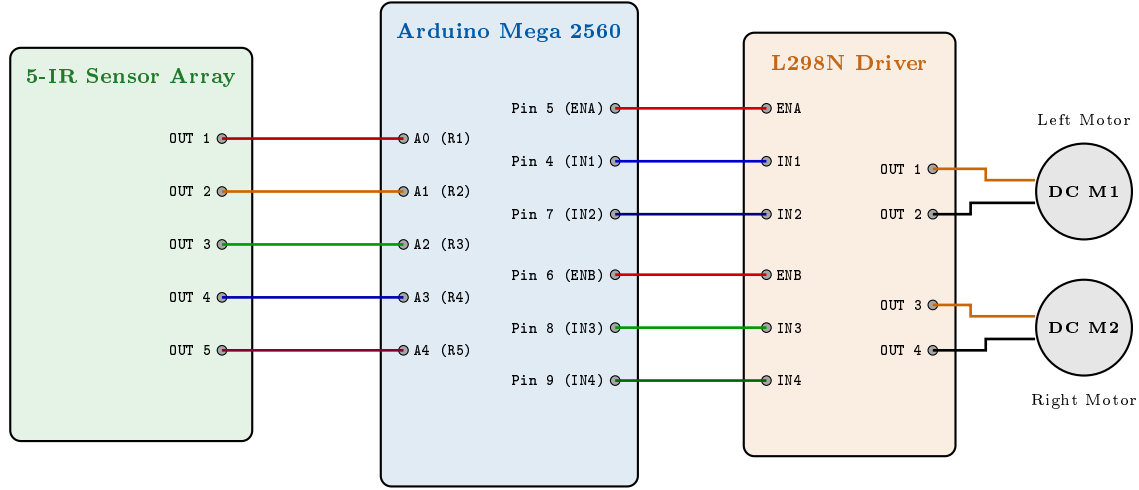


Figure 1. Circuit schematic of the line follower robot system.

The connections shown in Figure 1 establish signal routing and power distribution across the robotic system. The five optical outputs from the tracker board route to analog inputs A0 through A4 on the Arduino Mega 2560. Pulse width modulation pins 5 and 6 connect to the enable pins ENA and ENB of the L298N module to provide motor speed regulation. Digital pins 4, 7, 8, and 9 connect to inputs IN1 through IN4 to manage directional rotation. Output channels OUT1 and OUT2 energize the left motor, while channels OUT3 and OUT4 drive the right motor.

3 Logic Used and System Flowchart

The line following strategy uses five sensing points arranged symmetrically across the forward heading of the mobile platform. Sensor S1 occupies the far-left position, sensor S2 is on the inner left, sensor S3 occupies the true center, sensor S4 is on the inner right, and sensor S5 occupies the far-right position.

When a sensor hovers over the white floor, light reflects into the phototransistor, producing a low state of zero. When a sensor encounters the black trajectory, the light is absorbed, causing the sensor output to transition to a high state of one. The microcontroller samples all five inputs and determines the position of the line relative to the vehicle heading.

3.1 System Flowchart

Figure 2 shows the execution flow and decision branches governing the line follower robot.

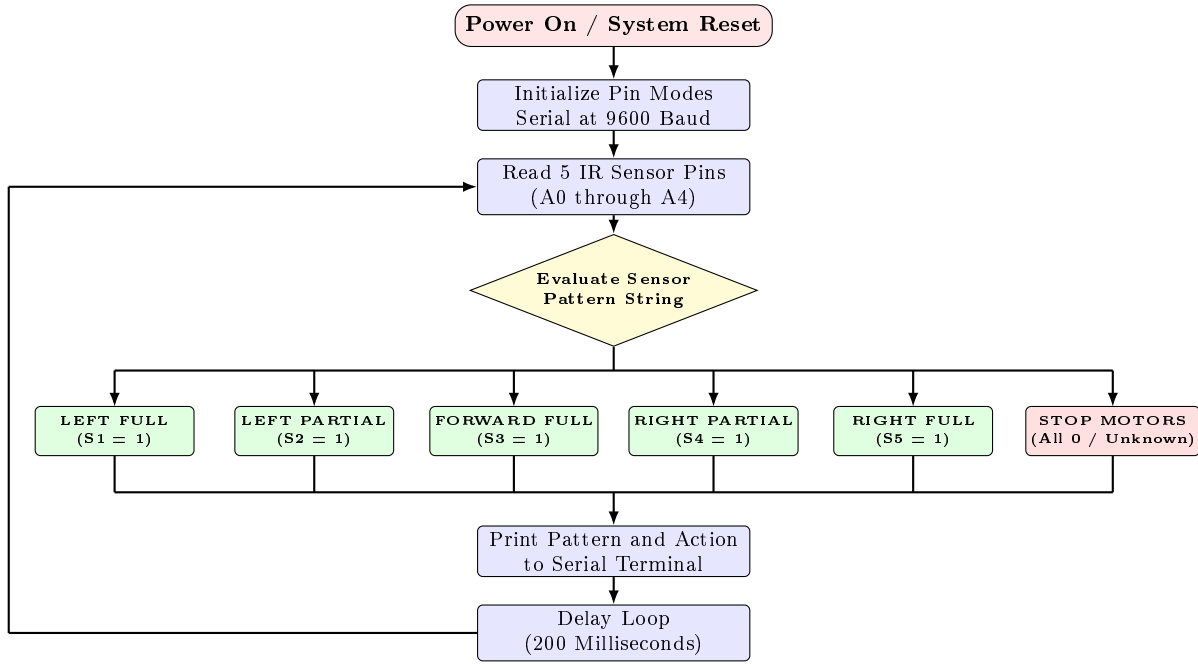


Figure 2. Flow diagram of the five-channel line tracking logic.

3.2 Algorithmic Structure or Pseudocode

Algorithm 1 details the sequential execution implemented in the firmware.

Algorithm 1. Decision algorithm for five-channel line tracking

```

1 Initialize serial communication at 9600 baud rate
2 Configure analog pins A0 through A4 as digital inputs
3 while true do
4   Read digital input from pin A0 into variable s1
5   Read digital input from pin A1 into variable s2
6   Read digital input from pin A2 into variable s3
7   Read digital input from pin A3 into variable s4
8   Read digital input from pin A4 into variable s5
9   Transmit all sensor values over the serial interface
10  if s1 == 0 and s2 == 0 and s3 == 0 and s4 == 0 and s5 == 0 then
11    | Print motor stop command
12  else if s3 == 1 and s2 == 0 and s4 == 0 then
13    | Print forward full speed command
14  else if s1 == 1 then
15    | Print turn left full speed command
16  else if s2 == 1 then
17    | Print turn left partial speed command
18  else if s5 == 1 then
19    | Print turn right full speed command
20  else if s4 == 1 then
21    | Print turn right partial speed command
22  else
23    | Print unknown state command
24  Wait for two hundred milliseconds
  
```

4 Code

The Arduino C++ implementation loaded onto the microcontroller board is given below.

```

1 #define S1 A0
2 #define S2 A1
3 #define S3 A2
4 #define S4 A3
5 #define S5 A4
6
7 void setup() {
8
9     Serial.begin(9600);
10
11     pinMode(S1, INPUT);
12     pinMode(S2, INPUT);
13     pinMode(S3, INPUT);
14     pinMode(S4, INPUT);
15     pinMode(S5, INPUT);
16 }
17
18 void loop() {
19
20     int s1 = digitalRead(S1);
21     int s2 = digitalRead(S2);
22     int s3 = digitalRead(S3);
23     int s4 = digitalRead(S4);
24     int s5 = digitalRead(S5);
25
26     // -----
27     // PRINT SENSOR VALUES
28     // -----
29
30     Serial.print("Sensors: ");
31
32     Serial.print(s1);
33     Serial.print(" ");
34
35     Serial.print(s2);
36     Serial.print(" ");
37
38     Serial.print(s3);
39     Serial.print(" ");
40
41     Serial.print(s4);
42     Serial.print(" ");
43
44     Serial.println(s5);
45
46
47     // -----
48     // DECISION LOGIC
49     // -----
50
51     // ALL WHITE
52     if (s1 == 0 &&
53         s2 == 0 &&
54         s3 == 0 &&
55         s4 == 0 &&
56         s5 == 0) {
57
58         Serial.println("ACTION: STOP ALL MOTORS");
59     }
60

```

```
61
62 // CENTER OF LINE
63 else if (s3 == 1 &&
64          s2 == 0 &&
65          s4 == 0) {
66
67     Serial.println("ACTION: MOVE FORWARD - FULL SPEED");
68 }
69
70
71 // LEFT FULL TURN
72 else if (s1 == 1) {
73
74     Serial.println("ACTION: TURN LEFT - FULL SPEED");
75 }
76
77
78 // LEFT PARTIAL TURN
79 else if (s2 == 1) {
80
81     Serial.println("ACTION: TURN LEFT - PARTIAL SPEED");
82 }
83
84
85 // RIGHT FULL TURN
86 else if (s5 == 1) {
87
88     Serial.println("ACTION: TURN RIGHT - FULL SPEED");
89 }
90
91
92 // RIGHT PARTIAL TURN
93 else if (s4 == 1) {
94
95     Serial.println("ACTION: TURN RIGHT - PARTIAL SPEED");
96 }
97
98
99 else {
100
101     Serial.println("ACTION: UNKNOWN");
102 }
103
104 Serial.println("-----");
105
106 delay(200);
107 }
```

5 Results

The integrated line follower system was tested under experimental floor tracking conditions. The five optical sensors continuously recorded ground reflectance, and the decision algorithm determined the required motor control actions.

Figure 3 displays the real-time serial monitor output obtained from the Arduino Mega 2560 operating at 9600 baud rate. When all five infrared sensors observe the high-reflectance white ground, the binary pattern reads 0 0 0 0 0, and the controller issues the command to stop all motors. When the tracking path shifts towards the rightmost detector, sensor S5 reads 1, and the system executes a full-speed right turn command to restore vehicle alignment over the guide trajectory. The experimental output confirms stable detection, prompt state transitions, and reliable decision execution throughout runtime.

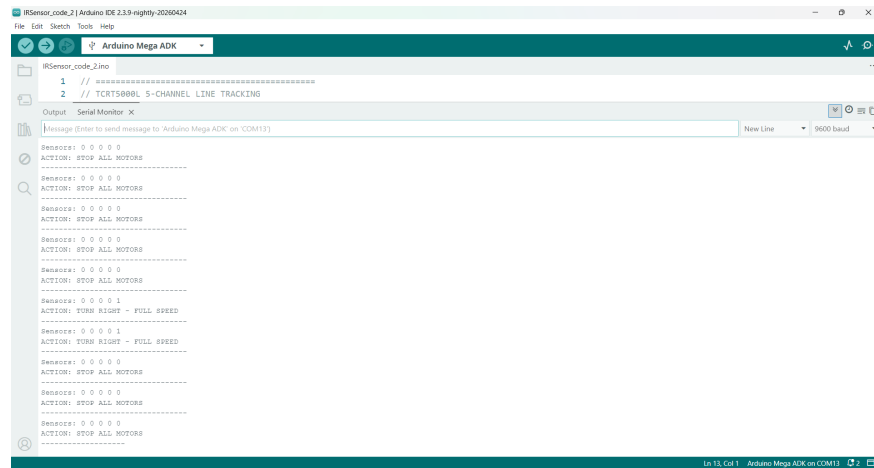


Figure 3. Serial monitor output captured during the infrared tracking experiment.

6 Video

The recorded experimental demonstration verifying the operation of the five-channel infrared sensor system is hosted on the institutional repository.

[Click Here to View the Infrared Sensor Demonstration Video](https://indianinstituteofscience.sharepoint.com/:v:/s/MN2072026-27Aug-Dec/IQDZuwr3FS9WQohTNFhevTPpAUhHFneh8fEJVC7A9Jka7LE?e=B9qfHJ)

The complete web link for manual browser access is provided below.

<https://indianinstituteofscience.sharepoint.com/:v:/s/MN2072026-27Aug-Dec/IQDZuwr3FS9WQohTNFhevTPpAUhHFneh8fEJVC7A9Jka7LE?e=B9qfHJ>